# Phase 5 - Carbon Accounting Engine

Status and required next inputs - CarbonHardware

This guide summarizes what is already implemented and what must still be supplied before the Phase 5 accounting outputs can be used in a research result.

## 1. Embodied carbon

The Phase 4 model can predict embodied carbon using process size, TDP, die area, die count, and average die area. Results are marked as model predictions, not measured LCA values. Use the saved artifact at ml-engine/artifacts/embodied_carbon_model.joblib.

**Yang perlu dilakukan:** Next input: hardware feature rows to predict, plus the model version, functional unit, and lifecycle boundary.

## 2. Operational carbon

Use operational_carbon = electricity_kwh * grid_intensity_kg_co2e_per_kwh. The engine does not contain a default grid value.

**Yang perlu dilakukan:** Next input: electricity consumption, grid intensity from an approved source, source URL or citation, analysis period, and assumptions such as utilization, TDP proxy, cooling, or PUE.

## 3. Compute Carbon Intensity (CCI)

Use CCI = total_carbon_kg_co2e / completed_compute_work.

**Yang perlu dilakukan:** Next input: an equivalent workload definition, for example completed training runs, inference requests, or tokens; also define lifecycle boundary and analysis period. Hardware comparisons must use equivalent workloads.

## 4. E-waste

The engine tracks physical mass in kg across reuse, recycling, and disposal. It intentionally keeps waste mass separate from carbon emissions.

**Yang perlu dilakukan:** Next input: hardware mass, end-of-life fractions, scenario source, replacement event, lifetime, and reuse/refurbishment assumptions. Fractions must sum to 1.

## 5. STEC

STEC is incomplete. The module refuses to produce a number because the verified spatial-temporal equation, dual attribution rule, required data, and correction factors are not available in the repository.

**Yang perlu dilakukan:** Next input: the complete cited paper, equations, spatial and temporal variables, allocation rule, units, and evidence for every correction factor. Do not invent missing factors.

## 6. Uncertainty

The current module summarizes source intervals or supplied samples. It does not claim probabilistic calibration or confidence coverage.

**Yang perlu dilakukan:** Next input: the project uncertainty framework, input distributions, parameter sources, model uncertainty definition, sample method, and calibration evidence if available.

## How to run the tests

From the project root, activate ml-engine/.venv and run: python -m unittest discover -s ml-engine/tests -p "test_*.py" -v

Phase 5 currently has no single dashboard or report runner. The modules are tested library functions. A future runner can combine the six result types after the required real inputs are supplied.

| Implemented | Needs research/input |
| --- | --- |
| Embodied prediction, operational formula, CCI formula, e-waste mass allocation, interval/sample uncertainty, validation tests. | Grid source/value, equivalent workload, hardware mass/routes, STEC equations and factors, uncertainty framework and calibration evidence. |